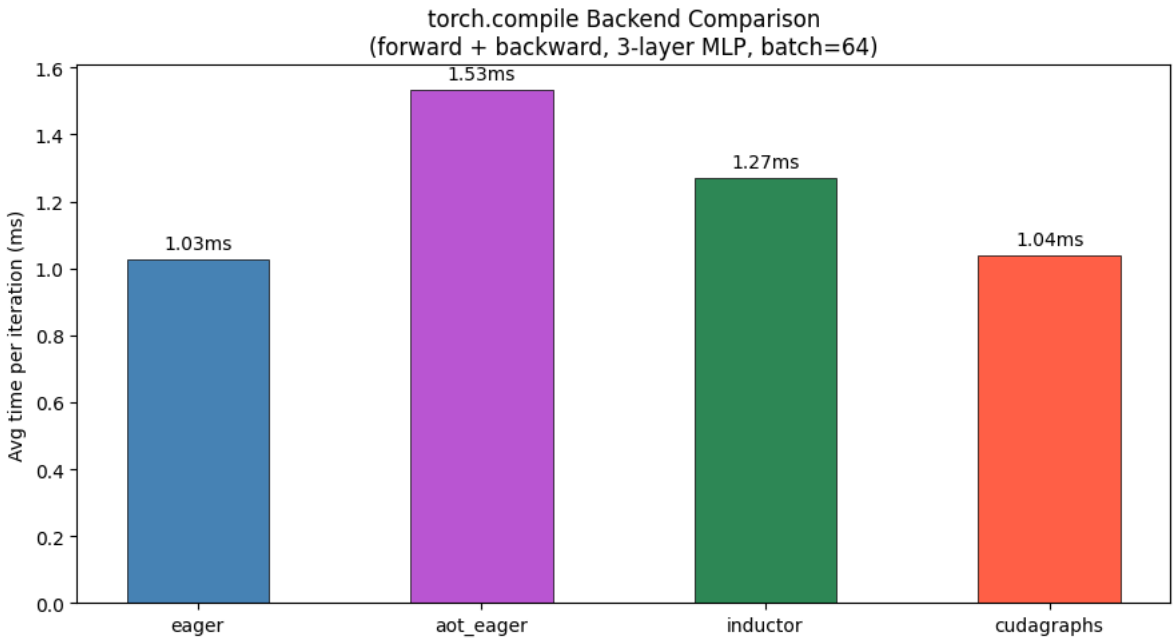
```
# Part 1: Comparing Backends (10 points)

import torch
import torch.nn as nn
import torch.nn.functional as F
import time
import matplotlib.pyplot as plt
import numpy as np

print('PyTorch version:', torch.__version__)
device = 'cuda' if torch.cuda.is_available() else 'cpu'
print('Device:', device)

class SimpleMLP(nn.Module):
    def __init__(self):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(512, 512),
            nn.ReLU(),
            nn.Linear(512, 512),
            nn.ReLU(),
            nn.Linear(512, 256),
        )

    def forward(self, x):
        return self.net(x)

backends = ['eager', 'aot_eager', 'inductor', 'cudagraphs']
N_RUNS = 100
BATCH_SIZE = 64
backend_times = {}

for backend in backends:
    model = SimpleMLP().to(device)
    optimizer = torch.optim.SGD(model.parameters(), lr=1e-3)

    if backend == 'eager':
        compiled_model = model
    else:
        compiled_model = torch.compile(model, backend=backend)

    x = torch.randn(BATCH_SIZE, 512, device=device)

    # Warmup — triggers compilation for compiled backends
    for _ in range(5):
        optimizer.zero_grad()
        out = compiled_model(x)
        loss = out.mean()
        loss.backward()
        optimizer.step()

    torch.cuda.synchronize()

    # Timed runs
    t0 = time.perf_counter()
    for _ in range(N_RUNS):
        optimizer.zero_grad()
        out = compiled_model(x)
        loss = out.mean()
        loss.backward()
        optimizer.step()

    torch.cuda.synchronize()

    avg_ms = (time.perf_counter() - t0) / N_RUNS * 1e3
    backend_times[backend] = avg_ms
    print(f'{backend:12s}: {avg_ms:.3f} ms/iter')

fig, ax = plt.subplots(figsize=(9, 5))
colors = ['steelblue', 'mediumorchid', 'seagreen', 'tomato']
names = list(backend_times.keys())
times = list(backend_times.values())

bars = ax.bar(names, times, color=colors, edgecolor='black', linewidth=0.5, width=0.5)
```

```
for bar, t in zip(bars, times):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.02,
            f'{t:.2f}ms', ha='center', va='bottom', fontsize=10)

ax.set_ylabel('Avg time per iteration (ms)')
ax.set_title('torch.compile Backend Comparison\n(forward + backward, 3-layer MLP, batch=64)')
plt.tight_layout()
plt.savefig('part1_backends.png', dpi=150)
plt.show()
```

```
PyTorch version: 2.10.0+cu128
Device: cuda
eager       : 1.028 ms/iter
aot_eager   : 1.532 ms/iter
W0317 00:08:15.603000 5892 torch/_inductor/utils.py:1679] [0/1] Not enough SMs to use max_autotune_gemm mode
inductor    : 1.269 ms/iter
cudagraphs  : 1.040 ms/iter
```




```
#  Explain: Why do different backends have different performance? What
# are the tradeoffs?

# **Results:**
# - eager:      1.028 ms/iter
# - aot_eager:  1.532 ms/iter
# - inductor:   1.269 ms/iter
# - cudagraphs: 1.040 ms/iter

# Eager runs operations immediately with no compilation overhead. Making it the fastest for small models. Where compilation cos
# AOT Eager traces the forward adn backward graphs ahead of time but performs no kernel optimization. It pays the tracing overh
# Inductor runs the full compilation pipeline. CoLab isn't able to fully leverage aggressive optimizations. Inductor typically
# CudaGraphs captures the entire CUDA op sequence as a static graph and replays it with a single driver call. Minimize kernel l

# Since the model is quite small,  eager wins out on being the fastest here. With a larger model that would not be the case.
```

```
# Part 2: Debugging Compilation Failures (10 points)

# --- BROKEN ---
def problem1(x):
    if x.sum() > 0:
        return x * 2
    else:
        return x / 2

# --- FIXED ---
def problem1_fixed(x):
    return torch.where(x.sum() > 0, x * 2, x / 2)

x = torch.randn(4, device=device)

# Test both
compiled_broken = torch.compile(problem1)
```

```
compiled_fixed  = torch.compile(problem1_fixed)

print('Broken output:', compiled_broken(x))
print('Fixed output: ', compiled_fixed(x))
print('Match:', torch.allclose(problem1(x), problem1_fixed(x)))

# Explain what Dynamo sees
print('\n--- Dynamo explain (broken) ---')
explanation = torch._dynamo.explain(problem1)(x)
print(explanation)
```

```
    Name: "L['x']"
    Source: local
    Create Function: TENSOR_MATCH
    Guard Types: ['TENSOR_MATCH']
    Code List: ["hasattr(L['x'], '_dynamo_dynamic_indices') == False"]
    Object Weakref: <weakref at 0x7947b491e610; to 'Tensor' at 0x7947b47dc9b0>
    Guarded Class Weakref: <weakref at 0x794914d18c70; to 'torch._C._TensorMeta' at 0x240f8e00 (Tensor)>
  Guard 9:
    Name: ''
    Source: shape_env
    Create Function: SHAPE_ENV
    Guard Types: None
    Code List: None
    Object Weakref: None
    Guarded Class Weakref: None
  Guard 10:
    Name: ''
    Source: global
    Create Function: DETERMINISTIC_ALGORITHMS
    Guard Types: None
    Code List: None
    Object Weakref: None
    Guarded Class Weakref: None
  Guard 11:
    Name: ''
    Source: global
    Create Function: GRAD_MODE
    Guard Types: None
    Code List: None
    Object Weakref: None
    Guarded Class Weakref: None
  Guard 12:
    Name: ''
    Source: global
    Create Function: DEFAULT_DEVICE
    Guard Types: ['DEFAULT_DEVICE']
    Code List: ['utils_device.CURRENT_DEVICE == None']
    Object Weakref: None
    Guarded Class Weakref: None
  Guard 13:
    Name: ''
    Source: global
    Create Function: GLOBAL_STATE
    Guard Types: None
    Code List: None
    Object Weakref: None
    Guarded Class Weakref: None
  Guard 14:
    Name: ''
    Source: global
    Create Function: TORCH_FUNCTION_STATE
    Guard Types: None
    Code List: None
    Object Weakref: None
    Guarded Class Weakref: None
  Guard 15:
    Name: ''
    Source: global
```

```
# --- BROKEN ---
def problem2(x):
    d = {}
    d['key'] = x
    return d['key'] * 2

# --- FIXED ---
def problem2_fixed(x):
    return x * 2

x = torch.randn(4, device=device)
```

```
compiled_broken = torch.compile(problem2)
compiled_fixed  = torch.compile(problem2_fixed)

print('Broken output:', compiled_broken(x))
print('Fixed output: ', compiled_fixed(x))
print('Match:', torch.allclose(problem2(x), problem2_fixed(x)))

print('\n--- Dynamo explain (broken) ---')
explanation = torch._dynamo.explain(problem2)(x)
print(explanation)
```

```
Broken output: tensor([-1.5310,  2.9746,  0.6986,  0.6846], device='cuda:0')
Fixed output:  tensor([-1.5310,  2.9746,  0.6986,  0.6846], device='cuda:0')
Match: True

--- Dynamo explain (broken) ---
Graph Count: 1
Graph Break Count: 0
Op Count: 1
Break Reasons:
Ops per Graph:
  Ops 1:
    <built-in function mul>
Out Guards:
  Guard 1:
    Name: ''
    Source: shape_env
    Create Function: SHAPE_ENV
    Guard Types: None
    Code List: None
    Object Weakref: None
    Guarded Class Weakref: None
  Guard 2:
    Name: ''
    Source: global
    Create Function: DETERMINISTIC_ALGORITHMS
    Guard Types: None
    Code List: None
    Object Weakref: None
    Guarded Class Weakref: None
  Guard 3:
    Name: ''
    Source: global
    Create Function: GRAD_MODE
    Guard Types: None
    Code List: None
    Object Weakref: None
    Guarded Class Weakref: None
  Guard 4:
    Name: ''
    Source: global
    Create Function: DEFAULT_DEVICE
    Guard Types: ['DEFAULT_DEVICE']
    Code List: ['utils_device.CURRENT_DEVICE == None']
    Object Weakref: None
    Guarded Class Weakref: None
  Guard 5:
    Name: ''
    Source: global
    Create Function: GLOBAL_STATE
    Guard Types: None
    Code List: None
    Object Weakref: None
    Guarded Class Weakref: None
  Guard 6:
    Name: ''
    Source: global
    Create Function: TORCH_FUNCTION_STATE
    Guard Types: None
```

```
# --- BROKEN ---
def problem3(x):
    result = 0
    for i in range(10):
        result += (x ** i).sum()
    return result

# --- FIXED ---
def problem3_fixed(x):
    powers = torch.stack([x ** i for i in range(10)])
    return powers.sum()
```

```
x = torch.randn(4, device=device)

compiled_broken = torch.compile(problem3)
compiled_fixed  = torch.compile(problem3_fixed)

print('Broken output:', compiled_broken(x))
print('Fixed output: ', compiled_fixed(x))
print('Match:', torch.allclose(
    torch.tensor(float(problem3(x))),
    problem3_fixed(x)
))

print('\n--- Dynamo explain (broken) ---')
explanation = torch._dynamo.explain(problem3)(x)
print(explanation)

print('\n--- Dynamo explain (fixed) ---')
explanation_fixed = torch._dynamo.explain(problem3_fixed)(x)
print(explanation_fixed)
```

```
Broken output: tensor(-12.4546, device='cuda:0')
Fixed output:  tensor(-12.4546, device='cuda:0')
Match: True

--- Dynamo explain (broken) ---
Graph Count: 1
Graph Break Count: 0
Op Count: 20
Break Reasons:
Ops per Graph:
  Ops 1:
    <built-in function pow>
    <built-in function add>
    <built-in function pow>
    <built-in function iadd>
    <built-in function pow>
    <built-in function iadd>
    <built-in function pow>
    <built-in function iadd>
    <built-in function pow>
    <built-in function iadd>
    <built-in function pow>
    <built-in function iadd>
    <built-in function pow>
    <built-in function iadd>
    <built-in function pow>
    <built-in function iadd>
    <built-in function pow>
    <built-in function iadd>
    <built-in function pow>
    <built-in function iadd>
Out Guards:
  Guard 1:
    Name: ''
    Source: shape_env
    Create Function: SHAPE_ENV
    Guard Types: None
    Code List: None
    Object Weakref: None
    Guarded Class Weakref: None
  Guard 2:
    Name: ''
    Source: global
    Create Function: DETERMINISTIC_ALGORITHMS
    Guard Types: None
    Code List: None
    Object Weakref: None
    Guarded Class Weakref: None
  Guard 3:
    Name: ''
    Source: global
    Create Function: GRAD_MODE
    Guard Types: None
    Code List: None
    Object Weakref: None
    Guarded Class Weakref: None
  Guard 4:
    Name: ''
```

```
# Part 3: Graph Capture and Inspection (15 points)

python_list = []  # external mutable state
```

```
def mixed_fn(x, W1, W2):
    # Two matrix multiplications
    h  = torch.mm(x, W1)
    h2 = torch.mm(h, W2)

    # ReLU activation
    h2 = torch.relu(h2)

    # Residual connection
    h2 = h2 + torch.mm(x, W2)

    # Layer normalization
    h2 = F.layer_norm(h2, h2.shape[-1:])

    # Python side effects — these should NOT be captured
    print('Hello from mixed_fn')
    python_list.append(h2.shape)

    return h2

x  = torch.randn(4, 8, device=device)
W1 = torch.randn(8, 8, device=device)
W2 = torch.randn(8, 8, device=device)

# fullgraph=True errors if there are any graph breaks
try:
    compiled_full = torch.compile(mixed_fn, fullgraph=True)
    out = compiled_full(x, W1, W2)
    print('fullgraph=True succeeded')
except Exception as e:
    print(f'fullgraph=True failed (expected): {type(e).__name__}')
    print(f'Reason: {e}')

compiled_partial = torch.compile(mixed_fn)
out = compiled_partial(x, W1, W2)
print('Output shape:', out.shape)
print('python_list contents:', python_list)

print('\n--- Dynamo explain ---')
explanation = torch._dynamo.explain(mixed_fn)(x, W1, W2)
print(explanation)

from torch._functorch.aot_autograd import make_fx

# Clean version with no side effects for make_fx
def clean_fn(x, W1, W2):
    h  = torch.mm(x, W1)
    h2 = torch.mm(h, W2)
    h2 = torch.relu(h2)
    h2 = h2 + torch.mm(x, W2)
    h2 = F.layer_norm(h2, h2.shape[-1:])
    return h2

fx_graph = make_fx(clean_fn)(x, W1, W2)

print('=== FX Graph IR ===')
fx_graph.print_readable()

print('\n=== Graph Nodes ===')
for node in fx_graph.graph.nodes:
    print(f'  op={node.op:12s}  name={node.name:30s}  target={str(node.target)}')
```

```
fullgraph=True failed (expected): Unsupported
Reason: Failed to trace builtin operator
  Explanation: Dynamo does not know how to trace builtin operator `print` with argument types ['str'] (has_kwargs False)
  Hint: Avoid calling builtin `print` with argument types ['str']. Consider using an equivalent alternative function/method to
  Hint: If you are attempting to call a logging function (e.g. `print`), you can try adding it to `torch._dynamo.config.reorde
  Hint: Please report an issue to PyTorch.

  Developer debug context: builtin print [<class 'torch._dynamo.variables.constant.ConstantVariable'>] False

 For more details about this graph break, please visit: https://meta-pytorch.github.io/compile-graph-break-site/gb/gb0059.html

from user code:
   File "/tmp/ipykernel_5892/2553727613.py", line 20, in mixed_fn
    print('Hello from mixed_fn')
```

```
Set TORCHDYNAMO_VERBOSE=1 for the internal stack trace (please do this especially if you're reporting a bug to PyTorch). For ev

Hello from mixed_fn
Output shape: torch.Size([4, 8])
python_list contents: [torch.Size([4, 8])]

--- Dynamo explain ---
Hello from mixed_fn
Graph Count: 1
Graph Break Count: 0
Op Count: 6
Break Reasons:
  Break Reason 1:
    Reason: Failed to trace builtin operator
  Explanation: Dynamo does not know how to trace builtin operator `print` with argument types ['str'] (has_kwargs False)
  Hint: Avoid calling builtin `print` with argument types ['str']. Consider using an equivalent alternative function/method to
  Hint: If you are attempting to call a logging function (e.g. `print`), you can try adding it to `torch._dynamo.config.reorde
  Hint: Please report an issue to PyTorch.

  Developer debug context: builtin print [<class 'torch._dynamo.variables.constant.ConstantVariable'>] False

 For more details about this graph break, please visit: https://meta-pytorch.github.io/compile-graph-break-site/gb/gb0059.html
    User Stack:
      <FrameSummary file /tmp/ipykernel_5892/2553727613.py, line 20 in mixed_fn>
Ops per Graph:
  Ops 1:
    <built-in method mm of type object at 0x79483e6e4b40>
    <built-in method mm of type object at 0x79483e6e4b40>
    <built-in method relu of type object at 0x79483e6e4b40>
    <built-in method mm of type object at 0x79483e6e4b40>
    <built-in function add>
    <function layer_norm at 0x7947e9c72840>
Out Guards:
  Guard 1:
    Name: ''
    Source: shape_env
    Create Function: SHAPE_ENV
    Guard Types: None
    Code List: None
    Object Weakref: None
    Guarded Class Weakref: None
  Guard 2:
```

```
# ## Computation Graph
#
#     x ──────────────────────────────┐
#     │                               │
#     ▼                               ▼
#   mm(x, W1)                     mm(x, W2)  ← residual branch
#     │                               │
#     ▼                               │
#   mm(h, W2)                         │
#     │                               │
#     ▼                               │
#   relu(h2)                          │
#     │                               │
#     └──────────── add ──────────────┘
#                   │
#                   ▼
#             layer_norm
#                   │
#                   ▼
#                output
#
# [ print() and python_list.append() — NOT in graph ]
```

```
## Part 3 Analysis

# **Which operations appear in the graph?**
# All pure tensor operations are captured (6 ops total):
# - torch.mm (×3 — two matmuls + one for the residual branch)
# - torch.relu
# - torch.add (residual connection)
# - native_layer_norm (decomposed internally into mean, variance, normalize)

# **Which operations are NOT captured?**
# - print('Hello from mixed_fn') — Python side effect with no tensor output.
#  fullgraph=True explicitly failed with: "Dynamo does not know how to trace
#  builtin operator print". Without fullgraph, Dynamo handles it by executing
#  the print in Python and continuing compilation after.
```

```
# - python_list.append(h2.shape) — External Python list mutation. Not a tensor
#  op so it never enters the graph. It executed once during tracing
#  (python_list contents: [torch.Size([4, 8])]) but will not re-execute
#  on subsequent compiled calls.

# **What happens to the print statement?**
# With fullgraph=True it causes a hard failure. Without fullgraph, Dynamo
# executes it eagerly during the traced call but does not include it in the
# compiled graph — it fires once at trace time and is then effectively gone.

# **What happens to the list modification?**
# Same story — executed once during tracing, never recorded in the graph.
# Subsequent compiled calls do not re-append to python_list. This is a
# common silent bug in compiled PyTorch code.
```